

K2-30b

R-2478

Benchmark run · workflow W8-benchmark-deep · record deep-bench_K2-30_201019 · created 2026-08-23T23:55:11+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459141.7463 +/- 0.0077 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.163 +/- 0.046
Transit depth (Rp/Rs)^2	2.67 +/- 1.5 [%]
Orbital Inclination (inc)	88.67 +/- 2.04
Airmass coefficient 1 (a1)	1.033 +/- 0.029
Airmass coefficient 2 (a2)	-0.012 +/- 0.013
Scatter in the residuals of the lightcurve fit is	2.79 %
Best Comparison Star	#1 - [283, 35]
Optimal Aperture	3.49
Optimal Annulus	11.25
Transit Duration (day)	0.062 +/- 0.013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.163 ± 0.046 vs 0.1266 (deviation 0.56σ)
Duration fit vs published	0.062 d vs 0.0971 d (deviation 1.89σ)
Mid-time O–C	16.1 min
Depth SNR	2.5
Residual scatter	2.79 %

Light curve

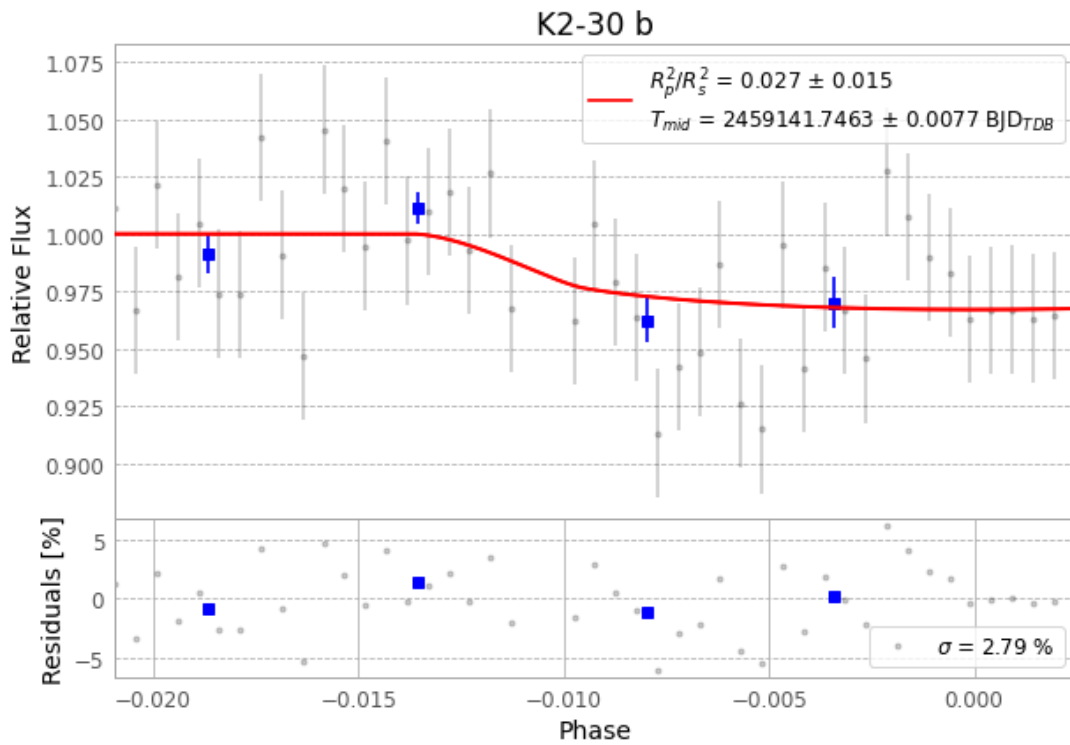


Figure 1 — FinalLightCurve_K2-30 b_2020-10-18.png (SHA-256 fe89cfb87604f113...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [346, 226], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 aca44ffeb8347909...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

45 FITS frames (30 MB), K2-30201019034212.FITS → K2-30201019060012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-K2-30-201019.log (2ece0eae2d9d...), FinalLightCurve_K2-30 b_2020-10-18.png (fe89cfb87604...), FinalLightCurve_K2-30 b_2020-10-18.csv (0eeea1736495...)

Compute resources

Wall time 92.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-23 23:55Z.